

Comparative Analysis for Peak Detection Techniques for Signal Processing

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- **Material:**
 - 25 noisy synthetic signals + Ground truth signals
 - Sampling frequency: 10 Hz
 - One example of a clean peak of intensity one

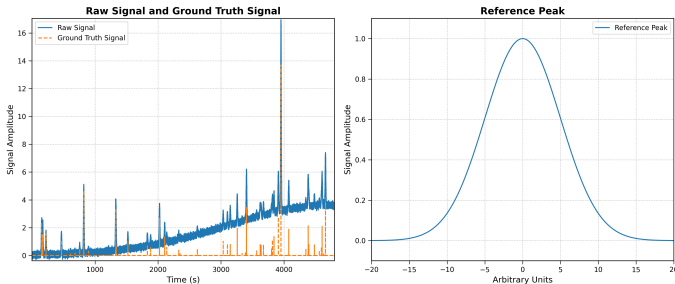


Figure: Noisy signal with ground truth peaks alongside the reference peak of intensity one.

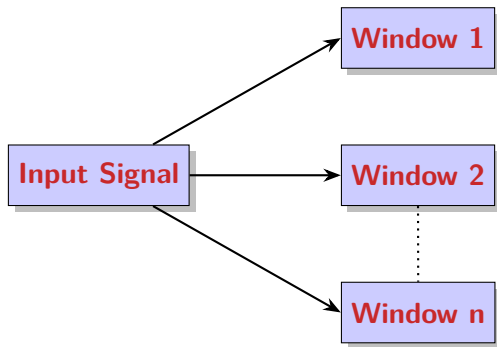
```
scipy_method(sig, t, gt_sig, fs=10, win_dur=500, th1=0.25, th2=0.15)
```

- **Parameters:**

- **sig, t:** Input signal and time vector.
- **gt_sig:** Ground truth signal for peak detection.
- **fs:** Sampling frequency (Hz).
- **win_dur:** Window duration in seconds.
- **th1:** Threshold factor for peak height.
- **th2:** Threshold factor for peak prominence.

- **Windowed approach:**

- **No ground truth peaks:** Skip the window.
- **One ground truth peak:** height = 25% max value, prominence = 15% max value.
- **Multiple ground truth peaks:** height = 25% max value, distance = average ground truth spacing $\times$ fs, prominence = 15% max value.



find_peaks(window_1, params)*

find_peaks(window_2, params)*

find_peaks(window_n, params)*

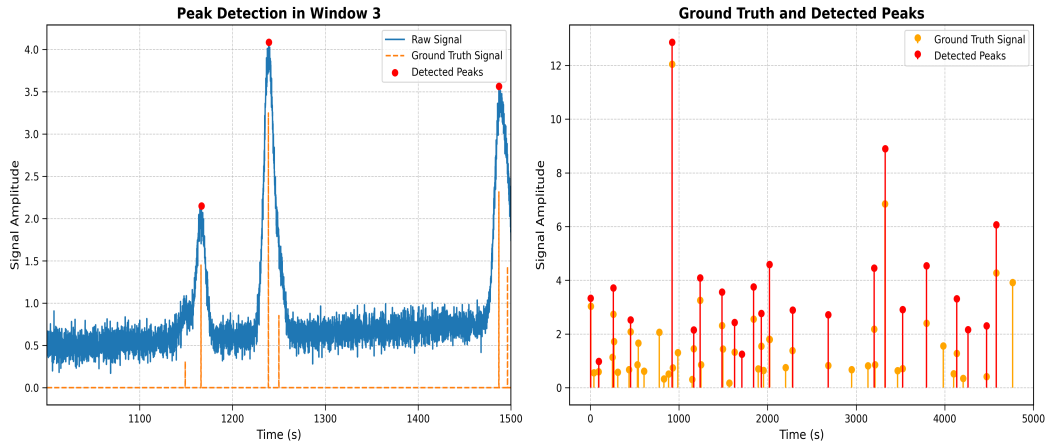
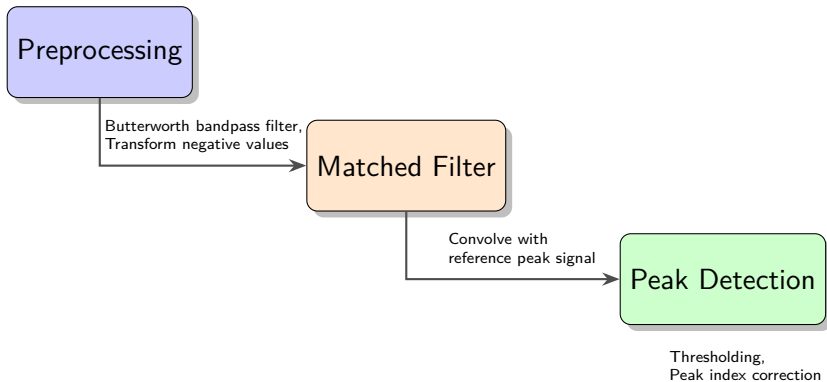


Figure: Peak detection process for a single window, along with the comparison of all detected peaks to ground truth peaks using the SciPy method.

```
hybrid_method(sig, t, ref, fs=10, order=1, lc=0.01, hc=0.1, th=0.01)
```

- **Parameters:**

- **sig**, **t**: Input signal and time vector.
- **ref**: Reference peak window for matched filtering.
- **fs**: Sampling frequency (Hz).
- **order**, **lc**, **hc**: Filter order, low cutoff frequency, and high cutoff frequency for the band-pass filter.
- **th**: Threshold factor to determine the peak detection threshold.



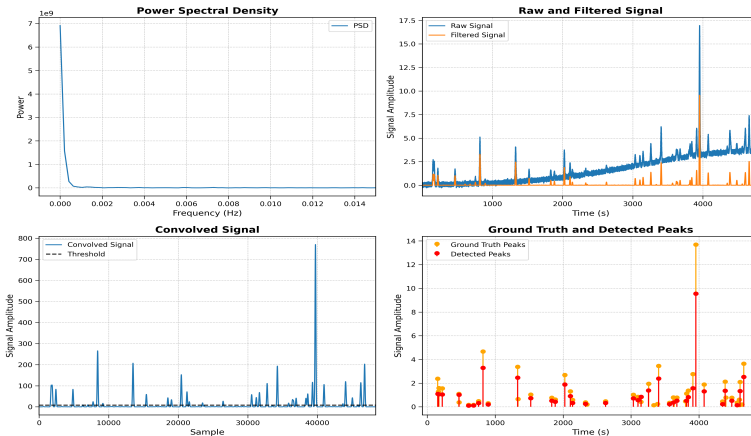
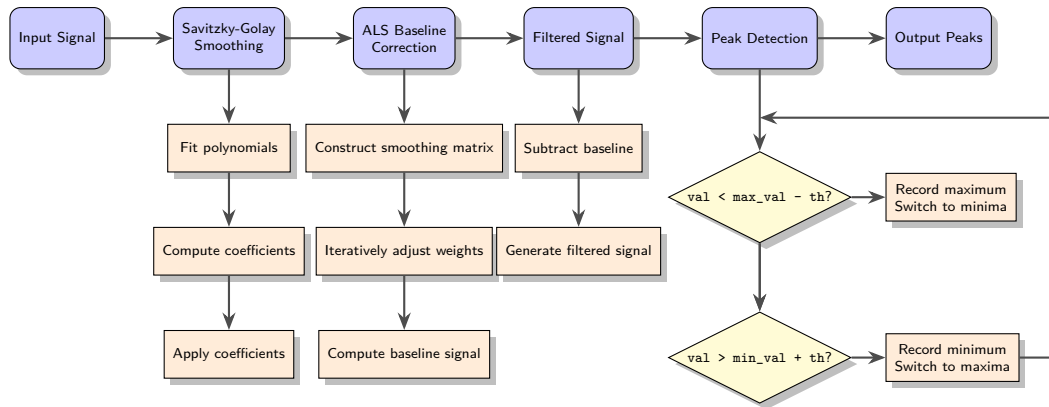


Figure: Representation of the power spectral density (PSD), raw signal, filtered signal, convolved signal, and the alignment of ground truth peaks with detected peaks using the hybrid method.

```
custom_method(sig, t, win_len=151, poly_order=3, lam=1e8, pen=0.001,  
              max_iter=50, th=0.1)
```

- **Parameters:**

- **sig, t:** Input signal and time vector.
- **win_len:** Window length for Savitzky-Golay smoothing.
- **poly_order:** Polynomial order for smoothing.
- **lam:** Smoothing parameter for ALS baseline removal.
- **pen:** Penalty parameter for ALS baseline removal.
- **max_iter:** Maximum iterations for ALS baseline removal.
- **th:** Threshold for peak detection.



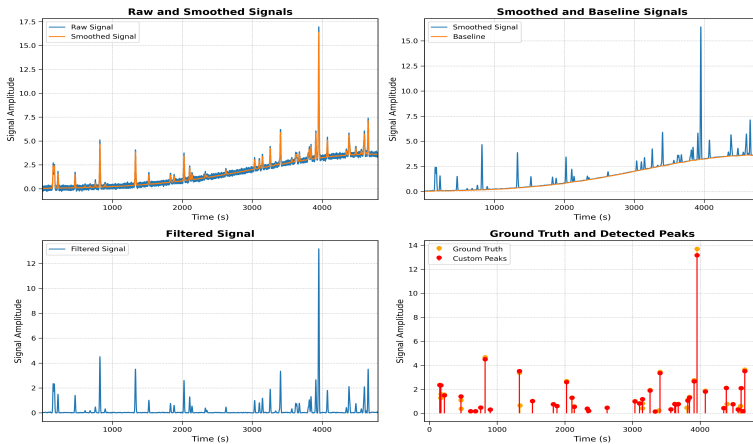


Figure: Raw, smoothed, baseline, and filtered signals processed using the custom method, along with detected peaks and ground truth peaks for comparison.

- **SciPy**: High variability, especially in specificity.
- **Hybrid**: Balanced performance, moderate variability.
- **Custom**: Consistent across all metrics.

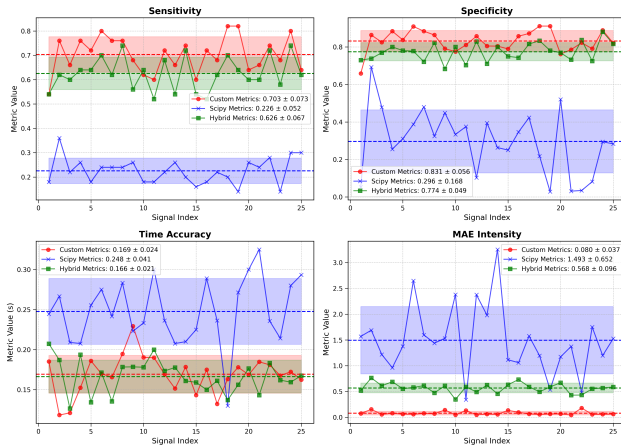


Figure: Performance comparison of all methods (tolerance = 500 ms).



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